

Symptoms and Their Temporal Distributions: An Interpretable AI Approach for Depression Detection in Social Media

Transparency Materials

A. Symptom Annotation Protocol and Instruction

1. Objective

This protocol aims to classify social media posts based on whether they contain content related to symptoms of depression, including both physical and psychological manifestations. The annotations will be used to support research on mental health and depression detection.

2. Post Labels and Examples

Depressive symptom	Category	Examples
Yes	Depressed mood	“The sky is as gray as my mood today. I feel exhausted even though I did nothing.” “I’ve been feeling really down lately. There’s no major problem in my life, but I just can’t seem to be happy...”
	Loss of pleasure or interest	“I used to love gaming, but now I don’t even want to log in.” “I don’t want to do anything, including living.”
	Poor concentration	“I have so much to do, but I just can’t focus. It’s driving me crazy.” “What was I about to do? I forgot as soon as I turned around. My memory is getting worse...”
	Feelings of excessive guilt or low self-worth	“Why do I always mess things up? Maybe I’m just not good enough.” “It’s all my fault. If it weren’t for me, they’d probably be happier...”
	Hopelessness about the future	“Feels like my life is stuck like this. I don’t see any hope for things to get better.” “As a kid, I dreamed about how bright the future would be. Now, all I see is darkness ahead.”
	Thoughts about dying or suicide	“Sometimes I think it wouldn’t be so bad if I just never woke up.” “Lately, I’ve been having some dark thoughts. What should I do...?”
	Disrupted sleep	“Haven’t slept well in a week. I’m a zombie during the day—I can’t take it anymore.” “3 AM again. My eyes are tired, but my brain won’t shut off.”
	Appetite disorder	“Haven’t eaten properly in three days. The sight of food makes me nauseous.” “Hungry, but don’t want to eat. Eating, but can’t taste anything.”
	Feeling very tired or low in energy.	“Did nothing, but I’m as exhausted as if I ran a marathon.” “Why am I still exhausted even after sleeping enough?”
	Others (Expert discretion)	“I don’t even feel sad anymore. Just...nothing.” “Why am I even like this. Everyone else seems to be okay, why not me.”
No	Neutral daily life updates	“Finally finished that project—time to relax!” “Weekend plans: Hiking with friends and maybe a movie night.”
	Positive emotions	“So grateful for my family’s support—couldn’t have done it without them!” “Laughing so hard with friends tonight. Needed this.”
	Factual posts	“New study shows walking 30 mins daily improves mental health.” “PSA: Metro Line 5 will be closed this weekend for maintenance.”
	Encouraging	“New month, new goals. Let’s make it count!” “Small steps still move you forward. Don’t give up!”
	Celebrations	“Happy birthday to my bestie! Love you!” “Graduation day! Four years of hard work paid off.”
	Others (Expert discretion)	“Just bought a new camera! Can’t wait to test it out.” “Learning guitar—my fingers hurt but it’s worth it.”

3. Annotation Instruction

Step 1: Read the Post Carefully

- Read the entire post without skimming to fully understand its context.
- Note the tone (e.g., sad, neutral, humorous, angry).
- Identify key phrases that may indicate emotional or mental state.

Step 2: Labeling Decision

- **“Yes” (Depressive Symptom-Related):** The post meets any of the following criteria:
 - ✓ **Explicit Symptoms:** Direct mentions of depressive symptoms (e.g., “I feel hopeless,” “I can’t get out of bed,” “Nothing matters anymore”).
 - ✓ **Implicit Indicators:** Metaphorical or behavioral hints of depression (e.g., “I’m drowning in my thoughts,” “I’ve been isolating myself for weeks”).
 - ✓ **Contextual Clues:** Posts where the overall tone and content strongly suggest depressive feelings, even if no single phrase is definitive.
- **“No” (Depressive Symptom-Unrelated):** Label the post as “No” when it does not meet the criteria for depressive symptom-related post.
- **Flag for Review:** Use this option when the post is ambiguous.

4. Quality Control

Dual Annotation: Each post is independently annotated by two annotators.

Consistency Calculation:

- Inter-annotator agreement (i.e., Cohen’s Kappa) is calculated to measure consistency.
- A minimum consistency score of **0.8** is required. If not met:
 - ✓ **Resolution:** The two annotators must discuss and reconcile differences.
 - ✓ **Re-annotation:** Low consistency (e.g., <0.7) may require re-annotation.

B. Symptom Annotation Process and Evaluation

1. Recruiting Process

We recruited annotators via direct email invitations sent to psychiatrists affiliated with well-known hospitals. Email addresses were obtained from publicly available hospital staff listings. A total of **15 invitation emails** were sent, and **5 responses** were received.

Two psychiatrists consented to do the annotation, each with more than 10 years of clinical practice and extensive experience diagnosing depression and related mood disorders. Both were provided with the Annotation Protocol and Instruction document and completed a qualification test consisting of 50 labeled examples. Both annotators achieved $\geq 95\%$ accuracy compared with gold-standard labels.

Recruitment Email

Subject: Invitation to Participate in Annotation of Depression-Related Social Media Posts

Dear Dr. [Surname],

We are conducting a research project aimed at identifying and categorizing depression-related symptoms expressed in social media posts. The purpose of this study is to improve the detection and understanding of depressive symptoms in online contexts, which may inform future clinical interventions and public mental health strategies.

We are seeking experienced psychiatrists to serve as expert annotators for a dataset of anonymized social media posts. The task will involve:

- Completing an initial qualification test (50 labeled examples)
- Reviewing posts and determining whether they describe or imply depressive symptoms.

Eligibility Criteria:

- Licensed psychiatrist with at least 10 years of clinical experience in psychiatry.
- Prior experience of diagnosing depression and related mood disorders.
- Willingness to annotate approximately 5,000 posts over a period of 2 weeks.

Compensation and Benefits:

- **Honorarium:** RMB ¥500 upon successful completion of the task.
- **Data Access:** Full access to the finalized annotated dataset for academic research.
- **Collaboration Opportunities:** Possibility to join our research group in future collaborative projects in mental health and computational psychiatry.

If you are interested in participating, please reply to this email within two weeks.

We look forward to the possibility of working together.

Best regards,

Research Team

2. Annotation Process and Evaluation

From the full dataset, 5,000 posts were randomly sampled. Each post was annotated independently by the two psychiatrists according to the Annotation Protocol and Instruction. Annotation was conducted using a shared Excel file containing post IDs, content, and columns for each annotator's label ("Yes" or "No").

We computed Cohen's Kappa to assess the inter-rater agreement before consensus discussion. The results are summarized below:

	Annotator B: Yes	Annotator B: No
Annotator A: Yes	295	8
Annotator A: No	10	4,687

- **Observed Agreement (Po)** = $(295 + 4,687) / 5,000 = 0.9964$
- **Expected Agreement (Pe)** = $[(305 \times 303) + (4,695 \times 4,697)] / (5,000^2) \approx 0.9936$
- **Cohen's Kappa** = $(Po - Pe) / (1 - Pe) \approx 0.93$

Following the initial annotation, the annotators reviewed the **18** posts with conflicting labels. Disagreements were resolved through discussion. The final dataset consisted of:

- **303** symptom-related posts
- **4,697** non-symptom-related posts

These finalized labels were then used for subsequent analysis in the study.

C. Survey Data Analysis

Table E1: Summary Statistics (Categorical)					
Variable	Category	Count	Variable	Category	Count
Age	0 - 17	0	Education	High school students and below	0
	18 – 24	28		Bachelor	28
	25 – 34	66		Master	102
	35 - 44	53		Doctor	35
	45+	18			
Gender	Female	86			
	Male	79			

Table E2: Summary Statistics (Continuous)						
Variable	Min	1 st Q	Median	Mean	3 rd Q	Max
Computer Literacy	1.00	2.00	3.00	3.02	4.00	4.00
Deep Learning Literacy	0.00	2.00	3.00	2.53	3.00	4.00
Trust in AI	0.00	2.00	3.00	2.66	3.00	4.00
Health Literacy	0.50	2.75	3.25	3.16	3.50	4.00
Depression Knowledge	0.20	0.60	0.80	0.71	0.80	1.00

Table E3: Randomization Checks								
	Age	Education	Gender	Computer Literacy	Deep Learning Literacy	Trust in AI	Medical Literacy	Depression Knowledge
P-value	0.154	0.267	0.488	0.623	0.798	0.290	0.426	0.840

Table E4. Correlation analysis on continuous variables in the user study						
	Computer Literacy	Deep Learning Literacy	Trust in AI	Medical Literacy	Depression Knowledge	Understandability
Computer Literacy	1.000	0.644	0.324	0.017	-0.064	0.104
Deep Learning Literacy	0.644	1.000	0.469	0.632	-0.076	0.180
Trust in AI	0.324	0.469	1.000	0.009	-0.134	0.249
Medical Literacy	0.017	0.063	0.009	1.000	0.004	0.175
Depression Knowledge	-0.064	-0.076	-0.134	0.004	1.000	-0.029
Understandability	0.104	0.180	0.249	0.175	-0.029	1.000

Table E5. Standardized Factor Loadings for the Understandability Construct					
Item	Estimate	Std. Error	z-value	p-value	Std. Loading
U1	1.000	–	–	–	0.712
U2	1.792	0.172	10.396	< 0.001	0.885
U3	1.342	0.144	9.306	< 0.001	0.780
U4	1.802	0.169	10.661	< 0.001	0.911
U5	1.689	0.172	9.834	< 0.001	0.830
U6	1.754	0.161	10.886	< 0.001	0.934
U7	1.602	0.157	10.196	< 0.001	0.865
U8	1.778	0.178	9.999	< 0.001	0.846

Reliability Assessment (R Language Implementation)

We calculated **Composite Reliability (CR)** using the standardized factor loadings:

```
CR <- (sum(std_loadings))^2 / ((sum(std_loadings))^2 + sum(1 - std_loadings^2))  
cat("CR =", round(CR, 3), "\n")  
# Output: CR = 0.953
```

The result indicates a high level of construct reliability (CR = 0.953).

We also computed **Cronbach's α** based on the item-level data (N = 165, k = 8 items):

```
item_variances <- apply(df, 2, var) # df: 165 × 8 (sample × items)  
total_variance <- var(rowSums(df))  
alpha_coef <- (k/(k-1)) * (1 - sum(item_variances)/total_variance) # k = 8  
print(alpha_coef)  
# Output: Cronbach's  $\alpha$  = 0.918
```

The obtained Cronbach's α further confirms excellent internal consistency reliability.

Table E6. Discriminant Validity		
	Medical literacy	Understandability
Medical literacy	0.725	
Understandability	0.122	0.849

D. Structured interview protocol

We invited six licensed medical professionals with clinical experience in depression screening and diagnosis. Each 30-minute interview was conducted online via video call with screen-sharing and recording (audio + screen). The interview consisted of two parts:

- **Background assessment.** Experts were first asked about their diagnostic experience and what they expect from an AI system that supports depression diagnosis.
- **Model evaluation.** For each expert, we sequentially presented a real case from the dataset and five corresponding model explanations, randomized in order. Each explanation included a predicted label (e.g., “likely depression”) along with model-specific interpretations. We encouraged think-aloud feedback and then asked structured questions:
 - ✓ **Interpretability:** “Can you understand how the model made this decision?”
Experts were asked to rank the five interpretations by interpretability and explain their reasoning.
 - ✓ **Clinical relevance:** “Does this explanation reflect key factors you consider when diagnosing depression?” “Would this explanation increase your confidence in the model’s prediction?” “Would this explanation help or hinder your diagnostic reasoning?”
 - ✓ **Potential usefulness for various stakeholders:** “Beyond your own clinical use, who else (e.g., patients, families) could benefit from such explanations, and in what ways?”